tc and IP fragments

Once defragmented, how to output them?

Marcelo Ricardo Leitner <marcelo.leitner@gmail.com>

The issue

- act_ct can do IP defrag
- But once that big packet is returned by it, tc (mirred) can’t output it
  - act_mirred has no knowledge today about IP fragments
    - The net device will reject it
  - No other action can output packets (not considering act_bpf)
- Issue noticed and patch proposed by wenxu:
  - https://lore.kernel.org/netdev/1593485646-14989-1-git-send-email-wenxu%40ucloud.cn/T/
  - **[PATCH net] net/sched: act_mirred: fix fragment the packet after defrag in act_ct**
    - Rejected by Cong Wang

Current state

- wenxu sent a new patch
  - https://lore.kernel.org/netdev/1596163501-7113-1-git-send-email-wenxu%40ucloud.cn/T/
  - **[PATCH net v2] net/sched: act_ct: fix miss set mru for ovs after defrag in act_ct**
  - It is enough for OVS, but only for it
    - Relies on a chain miss so that the packet is picked up by OVS and output using its kernel datapath
  - Still needs a tc-only fix

Sample of usage

tc filter add dev ens1f0_0 ingress prio 1 chain 0 proto ip flower ip_proto tcp ct_state -trk action ct nat pipe action goto chain 2

tc filter add dev ens1f0_0 ingress prio 1 chain 2 proto ip flower ct_state +trk+new action ct nat dst addr 3.3.3.42 commit pipe action mirred egress redirect dev ens1f0_1

tc filter add dev ens1f0_0 ingress prio 1 chain 2 proto ip flower ct_state +trk+est action mirred egress redirect dev ens1f0_1

Brainstorm: how tc can output it on its own?

- tc pipeline works with a single packet at a time:
  - act_ct can’t just return a list of packets
    - Multiple return codes in tcf_action_exec...
  - act_ct needs the reassembled packet in a second moment (nat on new entries)
- Similar to wenxu’s original approach: fix it in act_mirred, but only enable the feature if a flag says so
  - act_l3mirred ?
- (Ab)use interface backlog
  - Similar to reclassify, but doesn’t need to hold context
  - Adds latency (and reordering), but it’s IP frags anyway
- To be considered: ICMPs FragNeeded
- ?